

HOSEIN MOGHADDASI

Senior Exploration Geologist & Geophysicist | 3D Data Inversion Expert

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PROFESSIONAL SUMMARY

"Senior Exploration Professional with 14+ years of experience across the Tethyan Belt, combining field-based structural geology with advanced computational geophysics to maximize blind target discovery.

"I bridge the critical gap between structural geology and advanced mathematical geophysics to ensure that inversion models reflect drill-ready geological realities, not just mathematical artifacts. Specialized in geologically constrained 3D inversions using SimPEG (Python) and designing custom algorithms to pinpoint blind Porphyry, Porphyry Related, IOCG, and RIRGS systems beneath even alluvial cover. Proven track record of directing over 200 exploration projects.

CORE COMPETENCIES & TECHNICAL SKILLS

- **Geosciences:** Structural Geology, Alteration Interpretation, Target Generation, Mineral Prospectivity Mapping, Hydrogeology in Hard Rocks.
- **Computational Geophysics:** 3D Potential-Field Inversion, 2D/3D ERT & IP/Rs Modeling, Geologically Constrained Inversion, Legacy Data Upcycling, Fractional Derivatives, Wavelet Transforms.
- **Software & Algorithm Design:** SimPEG, Geo-Response Analyzer (In-house Algorithm Designer), Geosoft Oasis Montaj, ParaView, RES2DINV/RES3DINV, ArcGIS Pro.

PROFESSIONAL EXPERIENCE

CEO & Lead Geophysicist

Jarfabkavosh Consulting Engineering Co. | Tehran, Iran | 2013 – Present

Directed and executed 200+ diverse geophysical and geological exploration projects across Iran, ranging from complex metallic deposits to groundwater and geotechnical site investigations.

- **Advanced 3D Inversion:** Conducted complex 3D inversions of Total Magnetic Intensity (TMI) and Electrical data using SimPEG in Python. Utilized TreeMesh discretizations, Sparse, and L2-norm regularizations to accurately model subsurface high-susceptibility targets (e.g., Bahram Gur / Qatruyeh Project).
- **Blind Target Discovery & Drill Validation:** Successfully delineated a blind porphyry-related deposit covered by recent alluvium by applying Band pass filtering and also by Continuous Mexican Hat Wavelet transforms; target depth and mineralization were subsequently confirmed through confirmatory diamond drilling.
- **Integration:** Engineered custom Python workflows to invert scattered 1D Schlumberger sounding datasets into unified 3D models using SimPEG, drastically reducing 1D interpretation ambiguity.

- **QA/QC & Visualization:** Designed interactive Jupyter dashboards for real-time QA/QC, IGRF corrections, and iteration analysis of inversion results, exporting high-fidelity models to ParaView and Google Earth.
- **Field Operations:** Supervised extensive ground magnetic, IP, and ERT surveys for gold epithermal, skarn, and IOCG deposits across highly structurally deformed zones.

Senior Geophysical Advisor

Parman Pooyesh Co. | Iran | Jun 2026 – Present

Geophysical Advisor (Part-time)

Kowsar Mining & Industrial Development Investment Co. | Iran | Aug 2020 – May 2022

EDUCATION

Ph.D. Candidate in Geophysics (Goelectrical & Electromagnetics Methods)

Institute of Geophysics, University of Tehran, Iran | 2025 – 2028

M.Sc. in Geophysics (Goelectric)

Institute of Geophysics, University of Tehran, Iran | 2012 – 2015

- *GPA:* 16.54 / 20.00
- *Thesis:* Study of groundwater resources in hard rocks combining satellite imagery processing and various electrical methods (Resistivity, SP, IP) with different arrays.

B.Sc. in Geology

Faculty of Science, Kharazmi University, Tehran, Iran | 2003 – 2007

SELECTED PUBLICATIONS & LECTURES

- **Algorithm Development:** "Analysis of error-causing factors in 2D geoelectrical studies and providing a solution to minimize them." *Journal of Economic Geology* (2024).
- **Porphyry Exploration:** "Integrated use of analytical signal and upward continuation for exploration of porphyry deposits." *16th Symposium of Iranian Society of Economic Geology* (2024).
- **Keynote Speaker:** "Application of geophysical methods in the study and exploration of copper porphyry deposits." *Tabriz University* (2024).
- **Corporate Training Instructor:** "Integration of Exploration Data and Optimization of Drill Hole Targeting." Taught advanced masterclasses for *Iran Minerals Production and Supply Company (IMPASCO)* and *Parman Pooyesh Co.* (2026).